

Kenney Tran

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Education

- **Boston University** Boston, MA
Bachelor of Arts in Computer Science *Expected Graduation: 2026*
 - **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Systems, Fullstack Development, Machine Learning, Cloud Computing

Experience

- **ML Engineer** Boston, MA
BU Spark! *January 2026 – May 2026*
 - Reduced professor grading time from **25+ minutes to 3 minutes per student (8x faster)** at **\$0.04 per assessment** by architecting multi-model pipeline using **GPT-4o-mini, Gemini 2.5, and Claude Sonnet**
 - Saved **~25 hours per 80-student course** by designing RAG pipeline that retrieves lecture content as grading context, enabling evaluation of diagrams and images within PDFs
 - Improved grading accuracy by **15%** across assignment types by implementing rubric-based AI evaluation with few-shot calibration, eliminating grader-to-grader scoring variance
 - Built a highly available interface connecting a Python ML backend to an instructor-facing frontend, supporting **100+ student assessments** per grading cycle for BU MET

Technical Skills

- **Languages:** Python, Java, TypeScript/JavaScript, C, SQL
- **Frameworks & Libraries:** React, Next.js, Node.js, FastAPI, Spring Boot
- **Tools & Databases:** Git (GitLab/GitHub), Docker, PostgreSQL, Redis, Elasticsearch, Azure/GCP (AWS concepts)

Projects

- **Real-Time Campus Transit Tracker (Next.js, TypeScript, Redis, Google Maps API):**
 - **Architected a scalable backend for a transit platform** serving BU campus commuters, integrating live geospatial data feeds using **TypeScript**
 - **Reduced API latency to <100ms** and cut compute costs by **60%** by adding a **Redis caching layer**
 - **Set up CI/CD** with automated end-to-end testing and rollback support, enabling faster feature releases with minimal deployment risk
- **Fixed-Point Hardware Accelerator for Image Filtering (C++, Vitis HLS, Vivado, Xilinx Kria KV260):**
 - Built a Gaussian blur accelerator on the Xilinx Kria KV260 hitting **42x kernel speedup** (3.6 ms/frame at **100 Mpx/s**) with **53.4 dB PSNR** over an ARM Cortex-A53 baseline
 - Cut FPGA resources across 5 HLS iterations: BRAM **180 to 0**, all **40 DSPs eliminated** via power-of-two weight shifts, FF **-88%**, LUT **-60%**
 - Diagnosed AXI-Lite I/O bottleneck (**360K transactions/frame**, 633ms end-to-end) and proposed AXI-Stream/DMA migration as the architectural fix
- **Airbnb Price Prediction Model (Python, scikit-learn, pandas, XGBoost):**
 - Built a Python ML pipeline predicting prices across **76,333** NYC Airbnb listings with log-price transformation, neighborhood-grouped imputation, and feature engineering on **26 features**
 - Compared **9 models** (Linear, Lasso, Decision Tree, XGBoost, Random Forest, Neural Network) with **10-fold cross-validated** hyperparameter tuning
 - Random Forest hit **R² = 0.71** (RMSE = 0.12 on log price), up from **R² = 0.05** on raw linear baseline; outperformed XGBoost (R² = 0.66) and Neural Network (R² = 0.61)